B7 Graphs of functions

Name: ______________________

Class: ______________________

Date: ______________________

Time: **312 minutes**

Marks: **259 marks**

Comments:

Q1.

A market trader notices that daily sales are dependent on two variables:

number of hours, t, after the stall opens

total sales, x, in pounds since the stall opened.

The trader models the rate of sales as directly proportional to $\frac{8-t}{x}$

After two hours the rate of sales is £72 per hour and total sales are £336

(a) Show that

$$x\frac{dx}{dt} = 4032(8-t)$$

(3)

(b) Hence, show that

$$x^2 = 4032t(16 - t)$$

(3)

(c) The stall opens at 09.30.

(i) The trader closes the stall when the rate of sales falls below £24 per hour.

Using the results in parts (a) and (b), calculate the earliest time that the trader closes the stall.

(6)

(ii) Explain why the model used by the trader is not valid at 09.30.

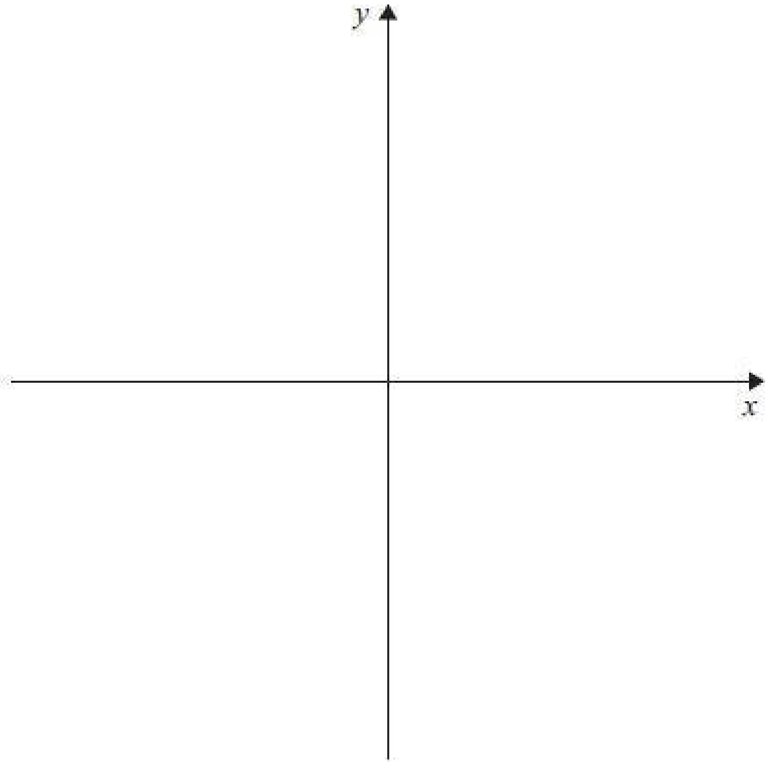
(2)

(Total 14 marks)

Q2.

Sketch the graph of $y = |2x + a|$, where a is a positive constant.

Show clearly where the graph intersects the axes.



(Total 3 marks)

Q3.

State the values of $|x|$ for which the binomial expansion of $(3 + 2x)^{-4}$ is valid.
Circle your answer.

$|x| < \frac{2}{3}$ $\qquad$ $|x| < 1$ $\qquad$ $|x| < \frac{3}{2}$ $\qquad$ $|x| < 3$

(Total 1 mark)

Q4.

The polynomial $p(x)$ is given by

$$p(x) = x^3 - 4x^2 - 3x + 18$$

(a) (i) Use the Factor Theorem to show that $x - 3$ is a factor of $p(x)$.

(2)

(ii) Express $p(x)$ as a product of linear factors.

(3)

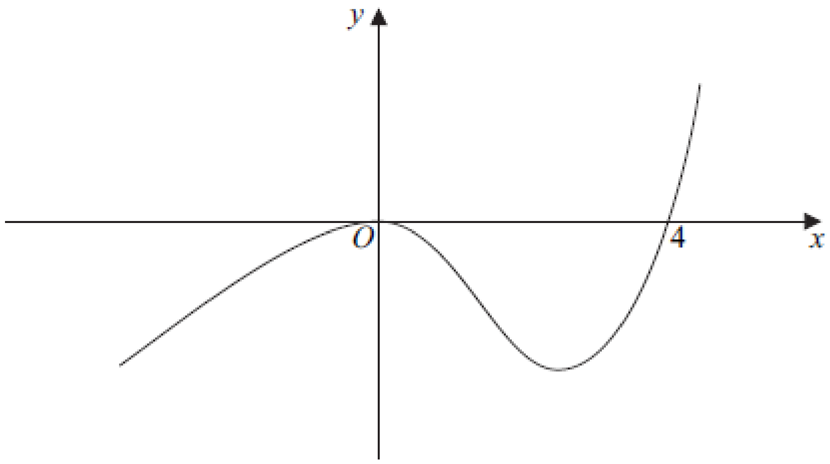
(b) Sketch the curve with equation $y = x^3 - 4x^2 - 3x + 18$, stating the values of x where the curve meets the x-axis.

(3)

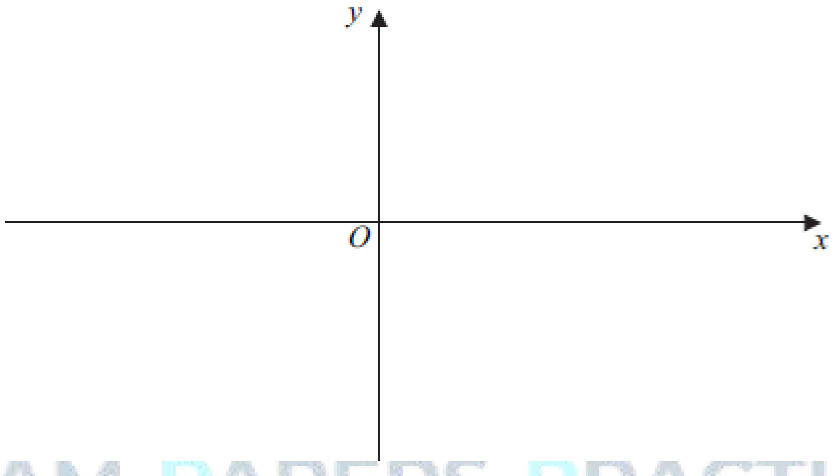
(Total 8 marks)

Q5.

The diagram shows a sketch of the curve with equation $y = \mathrm{f}(x)$.



(a) On the axes below, sketch the curve with equation $y = |\mathrm{f}(x)|$.



(2)

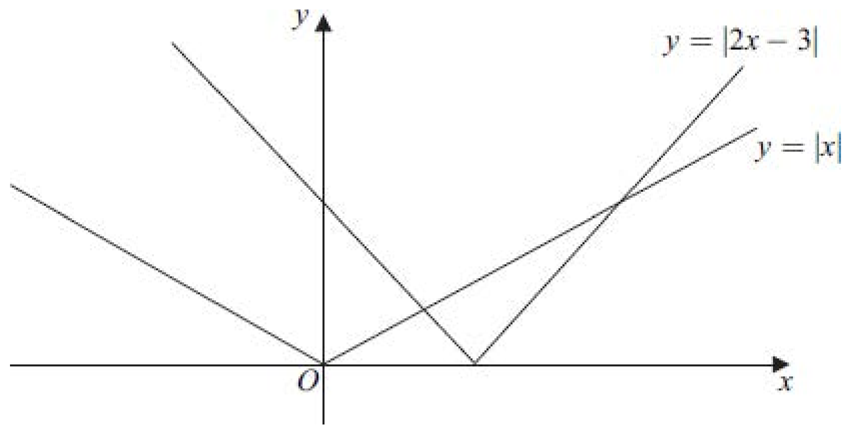
(b) Describe a sequence of two geometrical transformations that maps the graph of $y = \mathrm{f}(x)$ onto the graph of $y = \mathrm{f}(2x - 1)$.

(4)

(Total 6 marks)

Q6.

The diagram below shows the graphs of $y = |2x - 3|$ and $y = |x|$.



(a) Find the x-coordinates of the points of intersection of the graphs of $y = |2x - 3|$ and $y = |x|$.

(3)

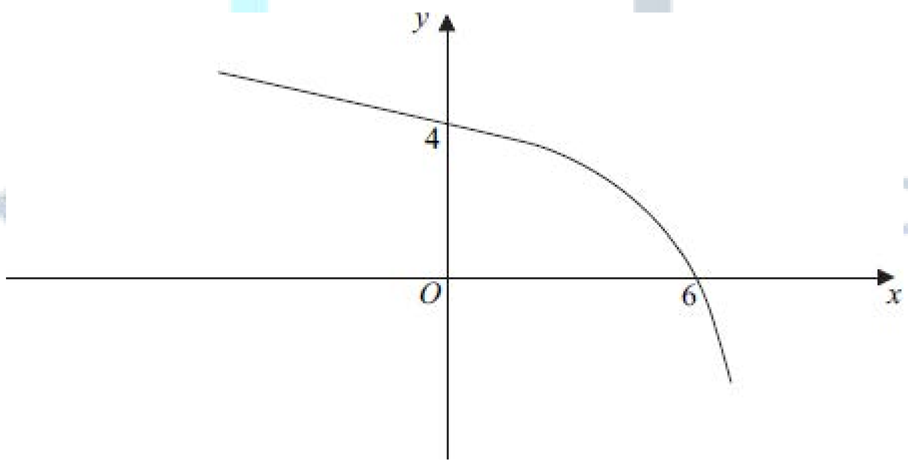
(b) Hence, or otherwise, solve the inequality

$$|2x - 3| \geq |x|$$

(2)
(Total 5 marks)

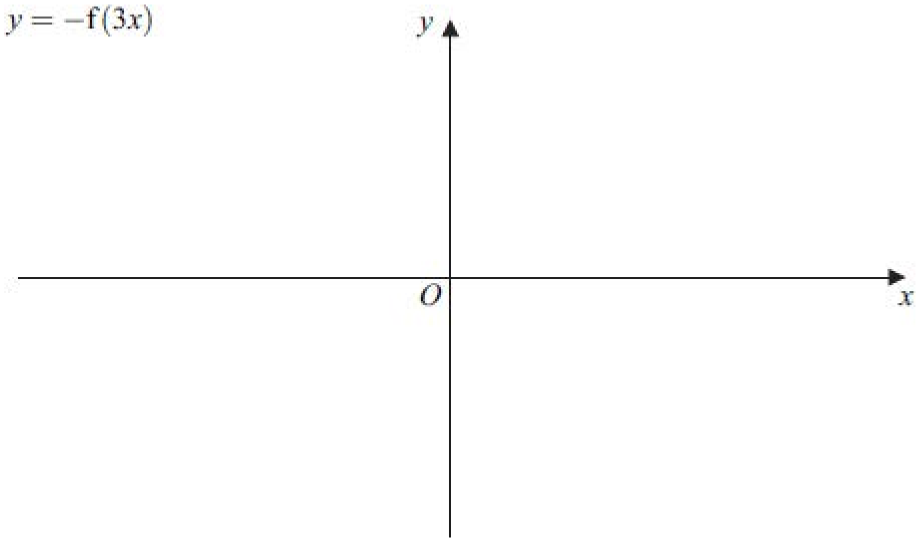
Q7.

The diagram shows a sketch of the curve with equation $y = f(x)$.



(a) On **Figure 1**, below, sketch the curve with equation $y = -f(3x)$, indicating the values where the curve cuts the coordinate axes.

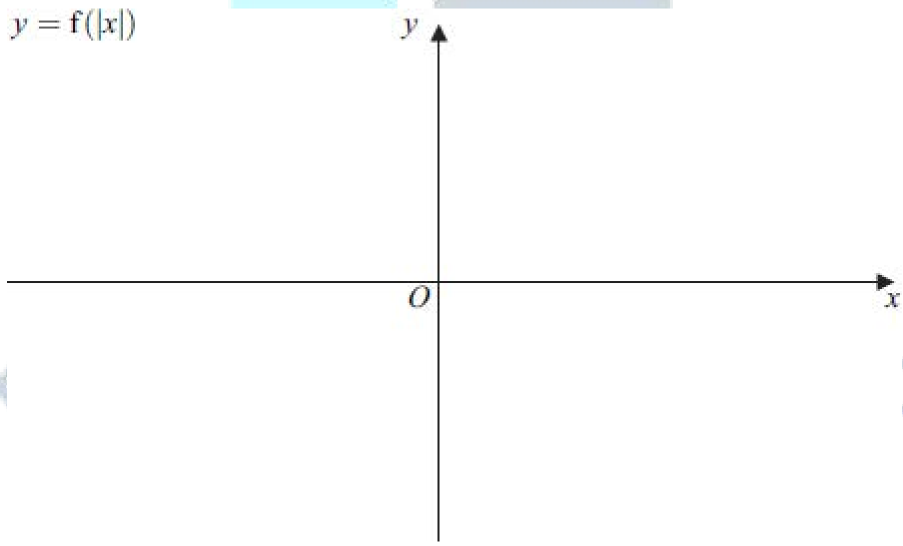
Figure 1



(2)

(b) On **Figure 2**, on the opposite page, sketch the curve with equation y = f($|x|$), indicating the values where the curve cuts the coordinate axes.

Figure 2



(3)

(c) Describe a sequence of two geometrical transformations that maps the graph of y = f(x) onto the graph of y = f$\left(-\frac{1}{2}x\right)$.

(4)

(Total 9 marks)

Q8.

The polynomial p(x) is given by

$$p(x) = x^3 + 2x^2 - 5x - 6$$

(a) (i) Use the Factor Theorem to show that $x + 1$ is a factor of $p(x)$.

(2)

(ii) Express $p(x)$ as the product of three linear factors.

(3)

(b) Verify that $p(0) > p(1)$.

(2)

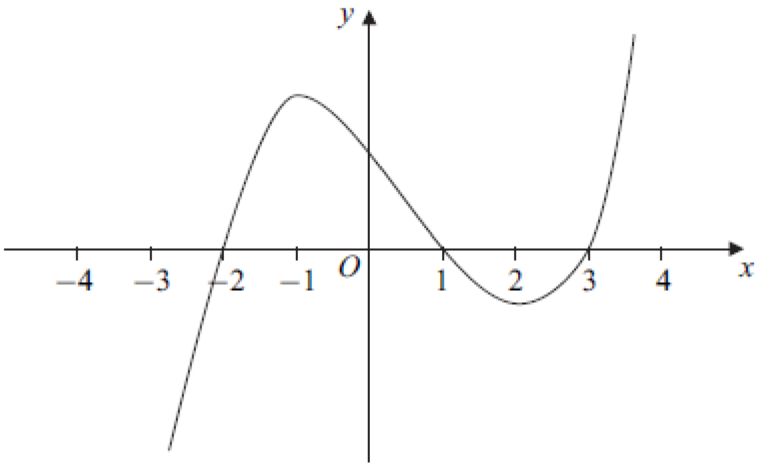
(c) Sketch the curve with equation $y = x^3 + 2x^2 - 5x - 6$, indicating the values where the curve crosses the x-axis.

(3)

(Total 10 marks)

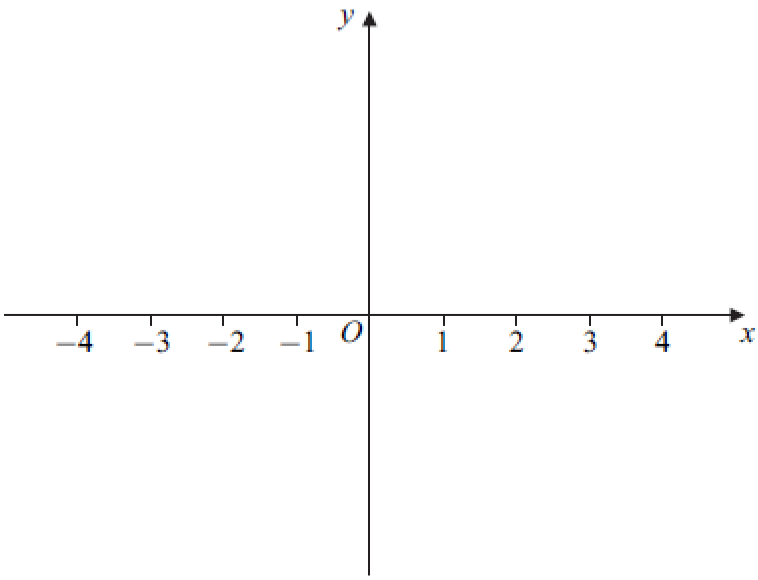
Q9.

The sketch shows part of the curve with equation $y = f(x)$.



(a) On **Figure 1** below, sketch the curve with equation $y = | f(x)|$.

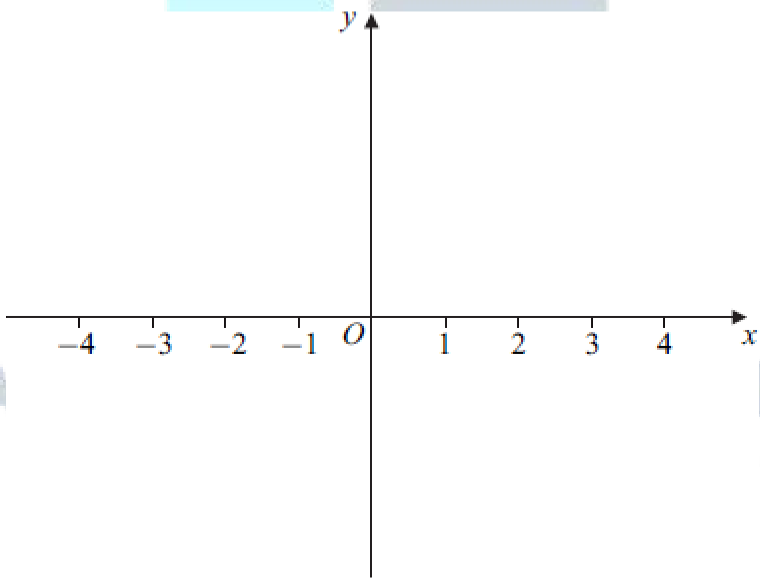
Figure 1



(3)

(b) On **Figure 2**, sketch the curve with equation $y = \mathrm{f}(\,|x|\,)$.

Figure 2



(2)

(c) Describe a sequence of two geometrical transformations that maps the graph of $y = \mathrm{f}(x)$ onto the graph of $y = \frac{1}{2}\mathrm{f}(x + 1)$.

(4)

(d) The maximum point of the curve with equation $y = \mathrm{f}(x)$ has coordinates (−1, 10).

Find the coordinates of the maximum point of the curve with equation $y = \frac{1}{2}\mathrm{f}(x + 1)$.

(2)

(Total 11 marks)

Q10.

The functions f and g are defined with their respective domains by

$\mathrm{f}(x) = 3\cos\frac{1}{2}x$,   for $0 \le x \le 2\pi$

$\mathrm{g}(x) = |x|$,   for all real values of x

(a) Find the range of f.

(2)

(b) The inverse of f is f^{-1}.

(i) Find $\mathrm{f}^{-1}(x)$.

(3)

(ii) Solve the equation $\mathrm{f}^{-1}(x) = 1$, giving your answer in an exact form.

(2)

(c) (i) Write down an expression for $\mathrm{gf}(x)$.

(1)

(ii) Sketch the graph of $y = \mathrm{gf}(x)$ for $0 \le x \le 2\pi$.

(3)

(d) Describe a sequence of two geometrical transformations that maps the graph of $y = \cos x$ onto the graph of $y = 3\cos\frac{1}{2}x$.

(3)

(Total 14 marks)

Q11.

(a) On separate diagrams:

(i) sketch the curve with equation $y = |3x + 3|$;

(2)

(ii) sketch the curve with equation $y = |x^2 - 1|$.

(3)

(b) (i) Solve the equation $|3x + 3| = |x^2 - 1|$.

(5)

(ii) Hence solve the inequality $|3x + 3| < |x^2 - 1|$.

(2)

(Total 12 marks)

Q12.

(a) Sketch the graph of $y = |8 - 2x|$.

(2)

(b) Solve the equation $|8 - 2x| = 4$.

(2)

(c) Solve the inequality $|8 - 2x| > 4$.

(2)

(Total 6 marks)

Q13.

The polynomial $p(x)$ is given by

$$p(x) = x^3 + 7x^2 + 7x - 15$$

(a) (i) Use the Factor Theorem to show that $x + 3$ is a factor of $p(x)$.

(2)

(ii) Express $p(x)$ as the product of three linear factors.

(3)

(b) (i) Verify that $p(-1) < p(0)$.

(1)

(ii) Sketch the curve with equation $y = x^3 + 7x^2 + 7x - 15$, indicating the values where the curve crosses the coordinate axes.

(4)

(Total 10 marks)

Q14.

(a) Solve the differential equation

$$\frac{dx}{dt} = -\frac{1}{5}(x + 1)^{\frac{1}{2}}$$

given that $x = 80$ when $t = 0$. Give your answer in the form $x = f(t)$.

(6)

(b) A fungus is spreading on the surface of a wall. The proportion of the wall that is unaffected after time t hours is x%. The rate of change of x is modelled by the differential equation

$$\frac{dx}{dt} = -\frac{1}{5}(x + 1)^{\frac{1}{2}}$$

At $t = 0$, the proportion of the wall that is unaffected is 80%. Find the proportion of the wall that will still be unaffected after 60 hours.

(2)

(c) A biologist proposes an alternative model for the rate at which the fungus is spreading on the wall. The total surface area of the wall is 9 m². The surface area that is **affected** at time t hours is A m². The biologist proposes that the rate of change of A is proportional to the product of the surface area that is affected and the surface area that is unaffected.

(i) Write down a differential equation for this model.

(You are not required to solve your differential equation.)

(2)

(ii) A solution of the differential equation for this model is given by

$$A = \frac{9}{1 + 4e^{-0.09t}}$$

Find the time taken for 50% of the area of the wall to be affected. Give your answer in hours to three significant figures.

(4)

(Total 14 marks)

Q15.

(a) The polynomial p(x) is given by p(x) = $x^3 - x + 6$.

(i) Use the Factor Theorem to show that $x + 2$ is a factor of p(x).

(2)

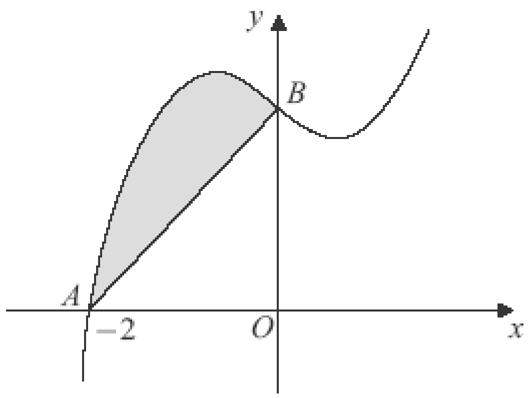
(ii) Express p(x) = $x^3 - x + 6$ in the form $(x + 2)(x^2 + bx + c)$, where b and c are integers.

(2)

(iii) The equation p(x) = 0 has one root equal to –2. Show that the equation has no other real roots.

(2)

(b) The curve with equation $y = x^3 - x + 6$ is sketched below.



The curve cuts the x-axis at the point $A(-2, 0)$ and the y-axis at the point B.

(i) State the y-coordinate of the point B.

(1)

(ii) Find $\int_{-2}^{0} (x^3 - x + 6)\,dx$.

(5)

(iii) Hence find the area of the shaded region bounded by the curve $y = x^3 - x + 6$ and the line AB.

(3)

(Total 15 marks)

Q16.

(a) The polynomial p(x) is given by p(x) = $x^3 - 7x - 6$.

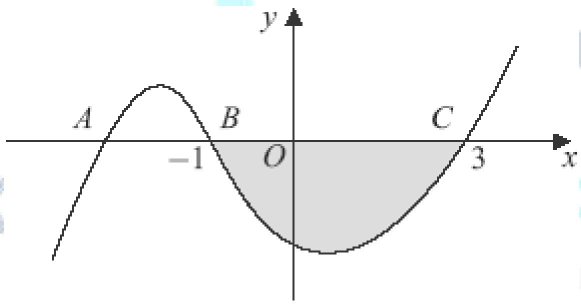
(i) Use the Factor Theorem to show that $x + 1$ is a factor of p(x).

(2)

(ii) Express p(x) = $x^3 - 7x - 6$ as the product of three linear factors.

(3)

(b) The curve with equation $y = x^3 - 7x - 6$ is sketched below.



The curve cuts the x-axis at the point A and the points $B(-1, 0)$ and $C(3, 0)$.

(i) State the coordinates of the point A.

(1)

(ii) Find $\int_{-1}^{3} (x^3 - 7x - 6)\,dx$.

(5)

(iii) Hence find the area of the shaded region bounded by the curve $y = x^3 - 7x - 6$ and the x-axis between B and C.

(1)

(iv) Find the gradient of the curve $y = x^3 - 7x - 6$ at the point B.

(3)

(v) Hence find an equation of the normal to the curve at the point B.

(3)

(Total 18 marks)

Q17.

(a) (i) Use the Factor Theorem to show that $x + 2$ is a factor of p(x).

(2)

(ii) Express p(x) as the product of linear factors.

(3)

(b) (i) The curve with equation $y = x^3 + x^2 - 8x - 12$ passes through the point $(0, k)$. State the value of k.

(1)

(ii) Sketch the graph of $y = x^3 + x^2 - 8x - 12$, indicating the values of x where the curve touches or crosses the x-axis.

(3)

(Total 9 marks)

Q18.

(a) The number of fish in a lake is decreasing. After t years, there are x fish in the lake. The rate of decrease of the number of fish is proportional to the number of fish currently in the lake.

(i) Formulate a differential equation, in the variables x and t and a constant of proportionality k, where $k > 0$, to model the rate at which the number of fish in the lake is decreasing.

(2)

(ii) At a certain time, there were 20 000 fish in the lake and the rate of decrease was 500 fish per year. Find the value of k.

(2)

(b) The equation

$$P = 2000 - Ae^{-0.05t}$$

is proposed as a model for the number of fish, P, in another lake, where t is the time in years and A is a positive constant.

On 1 January 2008, a biologist estimated that there were 700 fish in this lake.

(i) Taking 1 January 2008 as $t = 0$, find the value of A.

(1)

(ii) Hence find the year during which, according to this model, the number of fish in this lake will first exceed 1900.

(4)
(Total 9 marks)

Q19.

The polynomial $p(x)$ is given by

$p(x) = x^3 - 4x^2 - 7x + k$

where k is a constant.

(a) (i) Given that $x + 2$ is a factor of $p(x)$, show that $k = 10$.

(2)

(ii) Express $p(x)$ as the product of three linear factors.

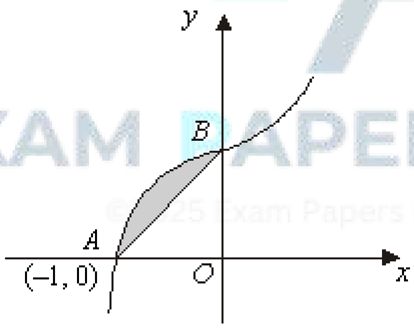
(3)

(b) Sketch the curve with equation $y = x^3 - 4x^2 - 7x + 10$, indicating the values where the curve crosses the x-axis and the y-axis. (You are **not** required to find the coordinates of the stationary points.)

(4)
(Total 9 marks)

Q20.

The curve with equation $y = 3x^5 + 2x + 5$ is sketched below.



The curve cuts the x-axis at the point A (–1, 0) and cuts the y-axis at the point B.

(a) (i) State the coordinates of the point B and hence find the area of the triangle AOB, where O is the origin.

(3)

(ii) Find $\int (3x^5 + 2x + 5)\,dx$

(3)

(iii) Hence find the area of the shaded region bounded by the curve and the line AB.

(4)

(b) (i) Find the gradient of the curve with equation $y = 3x^5 + 2x + 5$ at the point $A(-1, 0)$.

(3)

(ii) Hence find an equation of the tangent to the curve at the point A.

(1)
(Total 14 marks)

Q21.

(a) Sketch the graph of $y = |2x|$.

(1)

(b) On a separate diagram, sketch the graph of $y = 4 - |2x|$, indicating the coordinates of the points where the graph crosses the coordinate axes.

(3)

(c) Solve $4 - |2x| = x$

(3)

(d) Hence, or otherwise, solve the inequality $4 - |2x| > x$.

(2)
(Total 9 marks)

Q22.

The polynomial p(x) is given by

p(x) = $x^3 + x^2 - 10x + 8$

(a) (i) Using the factor theorem, show that $x - 2$ is a factor of p(x).

(2)

(ii) Hence express p(x) as the product of three linear factors.

(3)

(b) Sketch the curve with equation $y = x^3 + x^2 - 10x + 8$, showing the coordinates of the points where the curve cuts the axes.

(You are not required to calculate the coordinates of the stationary points.)

(4)
(Total 9 marks)

Q23.

(a) Sketch and label on the same set of axes the graphs of:

(i) $y = |x|$;

(1)

(ii) $y = |2x - 4|$.

(2)

(b) (i) Solve the equation $|x| = |2x - 4|$.

(3)

(ii) Hence, or otherwise, solve the inequality $|x| > |2x - 4|$.

(2)

(Total 8 marks)

Q24.

A disease is spreading through a colony of rabbits. There are 5000 rabbits in the colony. At time t hours, x is the number of rabbits infected. The rate of increase of the number of rabbits infected is proportional to the product of the number of rabbits infected and the number not yet infected.

(a) (i) Formulate a differential equation for $\frac{dx}{dt}$ in terms of the variables x and t and a constant of proportionality k.

(2)

(ii) Initially, 1000 rabbits are infected and the disease is spreading at a rate of 200 rabbits per hour. Find the value of the constant k.

(You are **not** required to solve your differential equation.)

(2)

(b) The solution of the differential equation in this model is

$$t = 4\ \ln\left(\frac{4x}{5000 - x}\right)$$

(i) Find the time after which 2500 rabbits will be infected, giving your answer in hours to one decimal place.

(2)

(ii) Find, according to this model, the number of rabbits infected after 30 hours.

(4)

(Total 10 marks)

Q25.

A car, of mass m, is moving along a straight smooth horizontal road. At time t, the car has speed v. As the car moves, it experiences a resistance force of magnitude $0.05mv$. No other horizontal force acts on the car.

(a) Show that

$$\frac{dv}{dt} = -0.05v$$

(1)

(b) When $t = 0$, the speed of the car is 20 m s^{-1}.

Show that $v = 20e^{-0.05t}$.

(4)

(c) Find the time taken for the speed of the car to reduce to 10 m s^{-1}.

(3)

(d) Find, in terms of m, the work done by the force in slowing the car from 20 m s^{-1} to 10 m s^{-1}.

(3)
(Total 11 marks)

Q26.

A car, of mass m, is moving along a straight smooth horizontal road. At time t, the car has speed v. As the car moves, it experiences a resistance force of magnitude $0.05mv$. No other horizontal force acts on the car.

(a) Show that

$$\frac{dv}{dt} = -0.05v$$

(1)

(b) When $t = 0$, the speed of the car is 20 m s^{-1}.

Show that $v = 20e^{-0.05t}$.

(4)

(c) Find the time taken for the speed of the car to reduce to 10 m s^{-1}.

(3)
(Total 8 marks)

Q27.

A sphere of mass 200 grams is released from rest and allowed to fall vertically.

(a) A student states that the acceleration of the sphere is 9.8 m s^{-2} while it is falling. What modelling assumption is this student making?

(1)

(b) The student conducts an experiment and finds that the acceleration of the ball is in fact 8 m s^{-2}. He formulates a model for the motion that assumes a constant resistance force acts on the ball as it is falling.

(i) Calculate the magnitude of this resistance force based on this assumption.

(3)

(ii) Describe how the resistance force would vary in reality.

(1)

(c) In a revised model the resistance force is assumed to be proportional to the speed of the sphere.

(i) State the initial acceleration of the sphere.

(1)

(ii) State what would happen to the acceleration of the sphere if it were able to fall for a long period of time.

(1)

(Total 7 marks)

least one mark. Not all candidates wrote the final answer as the product of three linear factors and were possibly confused by the repeated factor.

In part (b)(i), almost all candidates stated the value of k as –12, but a few wrote +12.

Most candidates found the graph in part (b)(ii) to be a difficult one to draw. Parabolas through (-2,0) and (3,0) were common and scored no marks. Those who drew a cubic shape usually had three intercepts on the x-axis and this was awarded one mark. Once again, the repeated root confused many. Some of those who made a good attempt lost a mark for having the minimum point on the y-axis.

Q18.

Many candidates seemed to have little experience of formulating a differential equation as often a derivative was missing from their answer to part (a)(i). Attempts at a derivative often involved variables other than the given x and t. Some candidates interpreted the question as 'find x in terms of t', and many expressions involving an exponential function were seen. Some candidates, who essentially had a correct differential equation, also thought they were required to solve it. Few candidates actually responded to the word 'decreasing' in the question, and many had a minus sign missing from an otherwise correct expression.

The given values of 500 and 20000 were often misinterpreted in part (a)(ii); the 500 often being taken to mean time. However, many candidates did indicate they had understood 500 to be a rate and 20000 the current value of x, but often also had a spurious t or another term in x in their differential equation. Some had a correct value of k, but negative, not apparently noting $k > 0$ had been given in the question. Relatively few fully correct answers to part (a) were seen.

Most candidates found the value of A correctly in part (b)(i), the common error being an apparent misinterpretation of the given information and finding A to be 700.

Part (b)(i) of the question was generally done well with many fully correct answers, and very few solutions by a trial and improvement approach. The majority of candidates could manipulate the expression and take logarithms correctly. Some algebraic errors were seen; in particular those who divided 1900 by 2000, rather than subtracting it, gained no further credit. Some first subtracted the 2000, and then tried to take logarithms of the expression as it stood, usually with an error. Some candidates incorrectly used base 10 logarithms. In completing the question most candidates rounded down their result to give the correct year. Some, who had made an error, did not round down and so had their year incorrect. Some candidates misinterpreted the last request in the question giving their answer as the 51st or 52nd year, rather than 2059.

Q19.

Part (a)(i) Most candidates found p(–2) but often failed to convince examiners that they had really shown that k = 10. Many substituted k = 10 from the outset and then drew no conclusion from the fact that p(–2) = 0. Those using long division often made sign errors.

Part (a)(ii) Factorisation of a cubic seems well understood and, apart from those who could not factorise $x^2 - 6x + 5$, candidates usually scored full marks. Some still confuse factors and roots.

Part (b) The sketch was generously marked with regard to the position of the stationary points but it was expected that candidates would indicate the values where the curve crossed the coordinate axes and often these values, particularly the 10 on the y-axis, were omitted.

Q20.

Part (a)(i) Some candidates ignored the request to state the coordinates of B even though they were using the height of the triangle as 5. The negative x-coordinate of A caused quite a few to feel that the triangle had a negative area. Far too many when finding $\frac{1}{2} \times 1 \times 5$ wrote the answer as 3.

Part (a)(ii) Practically every candidate found the correct integral although some made errors when cancelling fractions.

Part (a)(iii) It was necessary here to have the lower limit as –1 and the upper limit as 0. Many reversed the order and by some trickery arrived at a positive value. This was penalised and so very few, even though many had an answer of 1 for the area, scored full marks for this part of the question.

Part (b) Most candidates differentiated correctly but, because of poor understanding of negative signs, many wrong values of −13 were seen for the gradient. There is obviously confusion for many between tangents and normals and several thought the gradient of the tangent was $\frac{-1}{17}$.

Q21.

Parts (a) and (b) were usually well done, but in part (b), although many completely correct answers were seen, there were also many diagrams where part (a) had been translated 4 units along the x or y axis and W-shape diagrams were also popular. Dealing with the modulus in part (c) proved to be challenging, although there were many fully correct answers. Candidates appeared to be more successful with finding $x = \frac{4}{3}$ than x = –4. A surprising number of candidates followed 3x = 4 by $x = \frac{3}{4}$.

Those candidates who were successful in part (c) were usually successful in part (d) but correct values with incorrect inequality signs was a common error.

Q22.

(a) (i) It was good to see that almost all candidates started correctly by evaluating p(2), though a few thought they needed to find p(–2) and others wrongly assumed that long division was the “factor theorem”. It was necessary to write a conclusion or statement after showing that p(2) = 0 , in order to earn the second mark.

(ii) The most successful approach here was by using a quadratic factor ($ax^2 + bx$

+ c), though long division also worked well for many. A surprising number who found the correct quadratic then factorised it wrongly. Those who tried the factor theorem again rarely spotted both factors. A few lost the final mark by failing to write p(x) as a product of factors.

(b) Although there were many correct sketches, many lost a mark by failing to mark the point (0,8) on the y-axis. Candidates were expected to draw a cubic through their intercepts, to use an approximately linear scale and to continue the graph beyond the intercepts on the x-axis. It was common to see the negative values in the factors wrongly taken to be the roots and hence the intercepts on the x-axis.

Q23.

Part (a) Candidates scored well on this question with many correct graphs seen. Problems that occurred were in the gradients of the graphs being parallel or failure to label the points (2,0) and (0,4).

Part (b)(i) The most successful candidates on this question were the ones who used $x^2 = (2x - 4)^2$. The solution of $x = 4$ was a common response seen on its own. A surprising number followed $3x = 4$ by $x = 3/4$.

Part (b)(ii) The majority of candidates who obtained two values for x in part (i) were successful in obtaining the method mark for their extreme points but incorrect inequalities were seen.

Q24.

Most candidates struggled with part (a) of this question, indicating inexperience, or non understanding, of the formulation of a differential equation from a context situation.

Part (a) Despite the question explicitly requesting a differential equation for $\frac{\mathrm{d}x}{\mathrm{d}t}$ many candidates just wrote down an expression for x.

A commonly seen differential equation was $\mathrm{d}x/\mathrm{d}t = kx$ which candidates then proceeded to solve, despite the question stating that a solution of the differential equation was not required. Of those who could translate the given information into correct symbols, a symbol t for time, often found its way into their expression, and then often in an exponential form. Many attempts using the exponential function were seen, suggesting many candidates associating anything to do with differential equations with exponential forms. Those candidates who did give a correct differential equation in part (i) usually found a correct value for the constant of proportionality k. However, there were also many attempts from a partially correct part (i) where the given values of 1000 and 200 were confused, the latter often being taken as a value of time. Other candidates put forward the incorrect argument that the population would be equal to 1200 at time $t = 1$.

Part (b)(i) Virtually all candidates scored at least one mark here; the common error being to round off 5.545 to 5.6 or not to round it at all.

Part (b)(ii) Many candidates solved this equation correctly, although it was not always clear just how they had done it. Most candidates used the technique as per the mark scheme, although some worked successfully from $e^{30} = \left(= \frac{4x}{5000 - x}\right)^4$. Of those who went awry, the error usually involved an incorrect manipulation of the logarithm expression, including ignoring the ln altogether, or stating 4ln 4 = ln l6. Some started correctly with 7.5 = ln $4x$ – ln $(5000 – x)$ but obtained $e^{75} = 4x – (5000 – x)$. Others made relatively simple errors in signs or coefficients in their manipulation towards a solution.

Q25.

Parts (a), (b) and (c) were answered well. In part (d), few candidates used the fact that the work done by the force was equal to the change in kinetic energy. Most used complicated methods which rarely gave an appropriate answer.

Q26.

Virtually all candidates obtained $\frac{dv}{dt} = -0.05v$: only a few ignored the required step $m\frac{dv}{dt} = -0.05mv$. The equation $\int \frac{dv}{v} = -\int 0.05 \, dt$ was a necessary step which needed to be seen in part (b). Candidates knew roughly how to obtain $v = 20e^{-0.05t}$. Unfortunately, too often algebraic skills were not sufficient and the equation ln $v = -0.05 + c$ regularly became $v = e^{-0.05t} + c$ before becoming $v = Ae^{-0.05t}$. This and similar errors were not condoned.

Q27.

Good candidates tended to score well on this question while weaker candidates would only gain a few marks. Part (a) was often answered well, but candidates often missed the key assumption that was being made and stated assumptions that were not appropriate. In part (b) the most common error was to take the mass as 200 kg rather than 200 grams. When answering part (c), some candidates stated that the initial acceleration would be 8 ms^{-2} and some said that it would be zero. In the explanation, the quality of the expression was often poor and lacked clarity. Also some candidates did not explicitly state what would happen to the acceleration, but described what would happen to the velocity instead.